

Discrete Mathematics Quiz 2

Date: April 8, 2026, Time: 2.40 to 3.10

Write clear and concise solutions. Both problems carry equal marks.

1. How many distinct necklaces with 6 beads and 3 colors are there, where two necklaces are considered the same if one can be rotated and/or flipped to match the other? Derive the solution with a step by step explanation.
2. Let $G_1 \leq S_n$ and $G_2 \leq S_m$ be permutation groups acting on $[n]$ and $[m]$ with cycle index polynomials p_1 and p_2 , respectively. Derive the cycle index polynomial of their direct product $G_1 \times G_2$ in terms of p_1 and p_2 , where $G_1 \times G_2 \leq S_{nm}$ in its natural action on the set $[n] \times [m]$, which is $(a, b)^{(g_1, g_2)} = (a^{g_1}, b^{g_2})$.